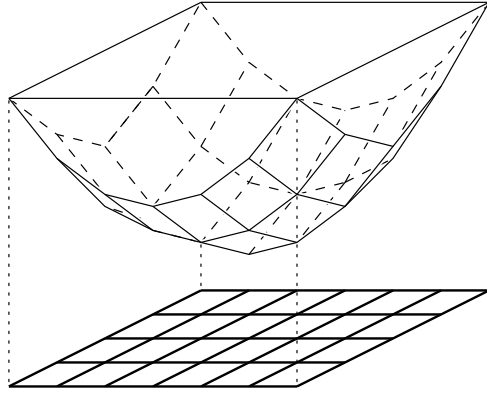
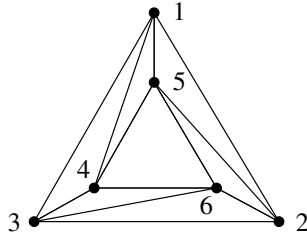


A canonical choice for f is the function $f(\mathbf{x}) = x_1^2 + \dots + x_d^2 = \|\mathbf{x}\|^2$. Our sketch tries to illustrate the construction of the 3-polytope $\mathcal{P}_3(6, 4)$.



To end this section, we give the following figure as a small example of a subdivision that is not regular. To see this, note that we can assume that $f(\mathbf{v}_4) = f(\mathbf{v}_5) = f(\mathbf{v}_6) = 0$ for the three interior vertices (by subtracting a linear function from f), and then we get a cycle $f(v_1) > f(v_2) > f(v_3) > f(v_1)$ of conditions for the three outer vertices.



Note that the subdivision $\Sigma_P(Q)$ encodes a lot of information about P , but essentially we “see only half of P ” in $\Sigma_P(Q)$. Now, we’ll “take a closer look” — in order to “see more.”

5.2 Schlegel Diagrams

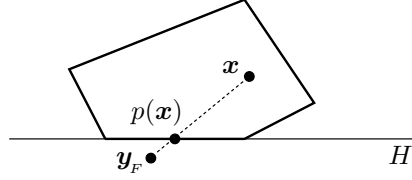
For a “close look” at polytopes, we choose a point of view $\mathbf{y}_F$ beyond facet F (see Section 3.1 for the definition of “beyond”) and use the facet F as a “projection screen” for everything we “see” behind it.

Definition 5.5. Let P be a d -polytope in $\mathbb{R}^d$, and let $F \in L(P)$ be a facet of P , defined by the valid inequality $\mathbf{a}\mathbf{x} \leq z$. We denote by

$$H := \text{aff}(F) = \{\mathbf{x} \in \mathbb{R}^d : \mathbf{a}\mathbf{x} = z\}$$

the hyperplane spanned by F . Choose a point $\mathbf{y}_F$ beyond F . For $\mathbf{x} \in P$, we define

$$p(\mathbf{x}) := \mathbf{y}_F + \frac{z - \mathbf{a}\mathbf{y}_F}{\mathbf{a}\mathbf{x} - \mathbf{a}\mathbf{y}_F}(\mathbf{x} - \mathbf{y}_F).$$



The *Schlegel diagram* of P based at the facet F , denoted as $\mathcal{D}(P, F)$, is the image under p of all proper faces of P other than F ; that is, it is the set system

$$\mathcal{D}(P, F) := \left\{ p(G) : G \in L(P) \setminus \{P, F\} \right\},$$

contained in the hyperplane H .

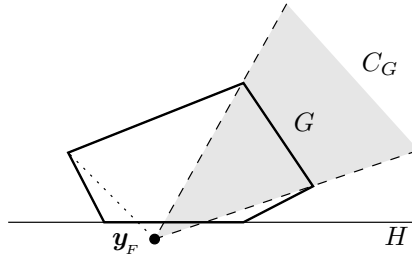
The map p is nonlinear, and in general we cannot replace “combinatorially equivalent” with “affinely isomorphic” in the following proposition; see Exercise 5.8. Now comes the reason why this construction makes sense.

Proposition 5.6. *The Schlegel diagram of P based at the facet F is a polytopal subdivision of F that is combinatorially equivalent to the complex $\mathcal{C}((\partial P) \setminus \{F\})$ of all proper faces of P other than F .*

Proof. We use the notation of Definition 5.5. For every face G of P , the set

$$C_G := \{\mathbf{y}_F + \lambda(\mathbf{x} - \mathbf{y}_F) : \mathbf{x} \in G, \lambda \geq 0\}$$

is a cone with vertex $\mathbf{y}_F$: in fact, $\mathbf{a}\mathbf{y}_F > \mathbf{a}\mathbf{x}$ for all $\mathbf{x} \in P$ by construction of $\mathbf{y}_F$.



If G is a proper face of P , then it is contained in a hyperplane

$$H_i = \{\mathbf{x} \in \mathbb{R}^d : \mathbf{a}_i \mathbf{x} = b_i\}$$

that does not contain $\mathbf{y}_F$. Thus we find that the face lattice of C_G is isomorphic to $L(G)$. Now if we intersect C_G with a hyperplane such as H , which has a bounded, nonempty intersection with C_G , then we have that

the face lattice of the intersection is isomorphic to that of C_G . Here we have $p(G) = C_G \cap H$ (the formula for $p(\mathbf{x})$ is derived from this condition), and thus $L(G) = L(C_G) = L(p(G))$.

(Actually, the map $p : G \rightarrow p(G)$ is a projective transformation, very much like the version we have in Section 2.6: see Exercise 5.8.) $\square$

Let us note here one very important property of Schlegel diagrams: that $G \cap \partial F$ is a face of $F = |\mathcal{D}|$, for all $G \in \mathcal{D}$.

Schlegel diagrams are, sure enough, named after a mathematician, Victor Schlegel [474]. It seems that Sommerville coined the name with his 1929 book [506]. In the next section we will get a more general definition of a d -diagram and then use “Schlegel” as an adjective for a d -diagram that is a Schlegel diagram.

If a Schlegel diagram of P is given as a polytopal complex $\mathcal{D}$, then we can reconstruct the corresponding facet F of P as $F = |\mathcal{D}|$. So with Proposition 5.6 we see that every Schlegel diagram $\mathcal{D}$ determines the combinatorial isomorphism type of P : we can reconstruct the face lattice of P from it. For this we recover $F = |\mathcal{D}|$, and get

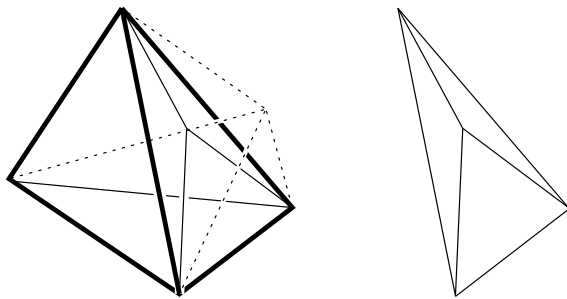
$$L(P) \cong (\mathcal{D} \cup \{F, \hat{1}\}, \leq),$$

where the partial order “ $\leq$ ” is by inclusion on $\mathcal{D}$, we have $G \leq F$ if and only if $G \in \mathcal{D}$ is a face of F , and $\hat{1}$ is an artificial maximal element.

The interesting property of the Schlegel diagram is that it completely encodes the combinatorial structure of a d -dimensional polytope into a $(d-1)$ -dimensional object; after all, $\mathcal{D}(P, F)$ is a polytopal complex contained in the $(d-1)$ -dimensional hyperplane $H = \text{aff}(F)$.

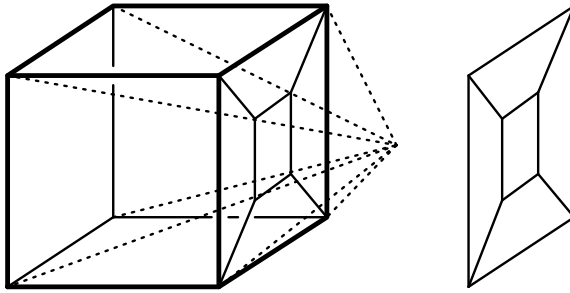
This reduction in dimension makes Schlegel diagrams especially useful in the case $d = 4$, where the Schlegel diagram of a 4-dimensional polytope is a 3-dimensional polytopal complex, for which we have a fighting chance for a geometric visualization.

To illustrate how Schlegel’s construction works, we start, however, at $d = 3$. The following drawings show the construction of the Schlegel diagram, and the (2-dimensional) Schlegel diagram itself, for a 3-dimensional simplex

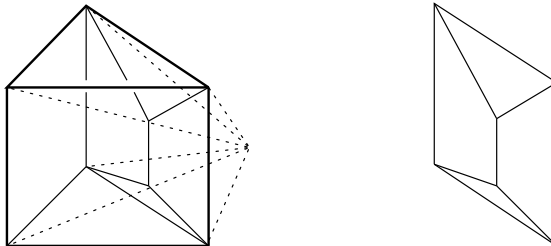
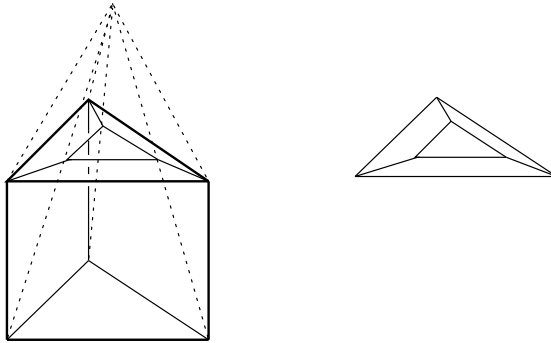


where the center vertex in the diagram represents the (unique) vertex of the tetrahedron which is not on the facet we project to. Similarly, here are

the drawings for the 3-dimensional cube.

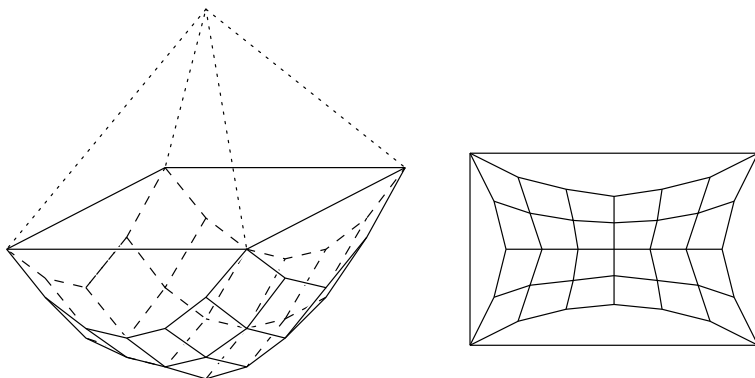


Next we consider the 3-dimensional prism (the product of a triangle with a 1-polytope): this has nonisomorphic facets, and thus we get different Schlegel diagrams for points of view beyond different facets (a square facet and a triangle facet).

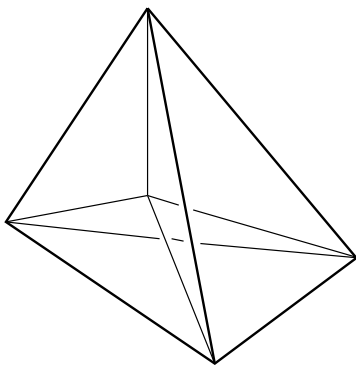


And finally, here is an attempt to illustrate how you construct the Schlegel diagram of a lifted pile of boxes, based on the big square facet on its top. The drawing is not metrically correct in any sense — just an attempt to understand the geometric situation. Specifically, we go for the Schlegel

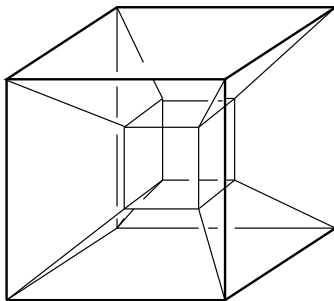
diagram of $\tilde{\mathcal{P}}_3(6, 4)$, from a point of view “above the polytope”:



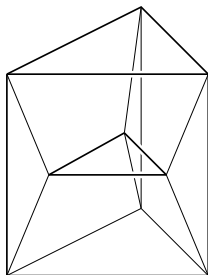
Now we go for the 3-dimensional Schlegel diagrams for 4-dimensional polytopes: these are all the “pictures” of 4-polytopes you will get! For example, here is (a 2-dimensional drawing of) the (3-dimensional) Schlegel diagram of the 4-dimensional simplex,



where the center vertex in the diagram represents the (unique) vertex of the 4-simplex which is not on the tetrahedron we project to; here is the drawing for the 4-cube,

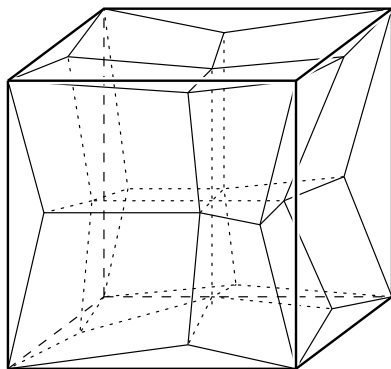


and here's the Schlegel diagram for a product of two simplices $P = \Delta_2 \times \Delta_2$,



a decomposition of a triangular prism into five triangular prisms — compare this to our description in Example 0.5.

We close with an attempted sketch of the Schlegel diagram of a 4-dimensional lifted pile of boxes, namely $\tilde{\mathcal{P}}_4(2, 2, 2)$.



Although this sketch is quite sketchy, the picture might try to tell you about the combinatorial structure of (the Schlegel diagram of) the 4-polytopes $\tilde{\mathcal{P}}_4(z_1, z_2, z_3)$. In fact, from the Schlegel diagram (compare also to the 3-dimensional case) we see that $\tilde{\mathcal{P}}_4(z_1, z_2, z_3)$ has exactly $z_1 z_2 z_3 + 7$ facets. The seven “big ones” are the big cubical facet on which the diagram is based and the six 3-dimensional facets that are lifted piles of cubes: two copies of $\tilde{\mathcal{P}}_2(z_1, z_2)$ at the bottom and the top of the diagram, two copies of $\tilde{\mathcal{P}}_2(z_1, z_3)$ at the front and the back of the diagram, and two copies of $\tilde{\mathcal{P}}_2(z_2, z_3)$ at the left and right sides of the Schlegel diagram. Between these six big facets, we have the subcomplex formed by $z_1 z_2 z_3$ little cubes, which is isomorphic to the pile $\mathcal{P}_3(z_1, z_2, z_3)$ from which we started the construction.

Make sure that you “see” this: it will be useful later (in Section 8.2) when we study the lifted piles of cubes again. More interesting structure connected with the “piles of cubes” polytopes constructed and studied here has recently been uncovered by Athanasiadis [25].

5.3 d -Diagrams

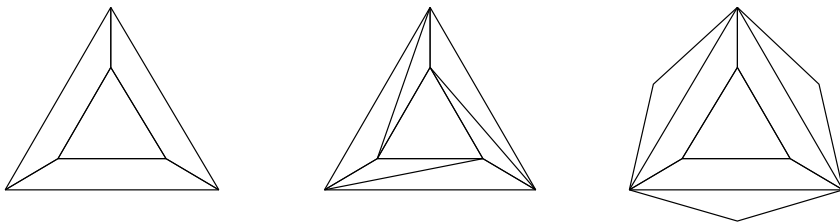
Schlegel diagrams were studied extensively around the turn of this century; however, no one realized that there is a problem: not everything that “looks like” a Schlegel diagram actually is one. (See the notes below.) Now we define *looks like*.

Definition 5.7. A d -*diagram* is a polytopal subdivision $\mathcal{D}$ of a d -polytope, $P = |\mathcal{D}| \subseteq \mathbb{R}^d$, such that $G \cap \partial P$ is a face of P for each $G \in \mathcal{D}$.

A d -diagram $\mathcal{D}$ is *simplicial* if $|\mathcal{D}|$ and all the polytopes in $\mathcal{D}$ are simplices. The diagram is *simple* if every vertex of $|\mathcal{D}|$ is contained in exactly d different d -polytopes of $\mathcal{D}$, and every other vertex of $\mathcal{D}$ is contained in exactly $d + 1$ different d -polytopes in $\mathcal{D}$.

Check that every Schlegel diagram of a d -polytope is a $(d - 1)$ -diagram! We define the face poset $L(\mathcal{D})$, and combinatorial equivalence, of d -diagrams as a special case of the definitions for polyhedral complexes. With this, a Schlegel $(d - 1)$ -diagram $\mathcal{D}(P, F)$ is simplicial, respectively simple (according to Definition 5.7), if and only if the polytope P is simplicial, respectively simple. In fact, a diagram $\mathcal{D}$ is simplicial if and only if its face lattice $\hat{L}(\mathcal{D})$ “looks like” the face lattice of a simplicial polytope, in that all the lower intervals $[\emptyset, G]$ are boolean for $G \neq \hat{1}$; similarly, $\mathcal{D}$ is simple if and only if all upper intervals $[G, \hat{1}]$ are boolean for $G \neq \emptyset$.

Here are a few trivial examples: a Schlegel diagram (of a triangular prism), a 2-diagram that is not a Schlegel diagram, and a polytopal subdivision that is not a 2-diagram.



We note that the question of whether a given d -diagram is a Schlegel diagram can be reduced to a linear programming problem (Exercise 5.2) — but the question of whether it is *combinatorially equivalent to a Schlegel diagram* is much harder; the task can be split into the enumeration problem for oriented matroid spheres, and the oriented matroid realizability problem (see the notes at the end of this lecture). So, the following is a decidedly nontrivial theorem, accompanied by a trivial proof.

Theorem 5.8.

Every 2-diagram is combinatorially equivalent to a Schlegel diagram.